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QUERY PROCESSING

1. CUSTOMER DATA CLEANING AND STANDARDIZATION

import pandas as pd
data = {

"customer_ID" : [101, 102, 103, 101, 104],

"Name" : ["John", "Alice", "Bob", "John",
"David"],

"City" : ["chennai ", "Chennai ", "CHENNAI",
"Chennai ", "Madurai"],
}

df = pd.DataFrame(data)

print("Original Dataset:")

print(df)

df = df.drop_duplicates()

df["City"] = df["City"].str.strip()

df["Phone"] = df["Phone"].fillna("Not available")

print("\n Clean Customer Master Dataset:")

print(df)

2. EMPLOYEE PAYROLL DATA

PREPARATION:

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103, 101, 104],
Bob, "John",
"David",
"CHENNAI",
", "Madurai"]

()

"Not available")
dataset: ")

```
import pandas as pd
import numpy as np
df = pd.DataFrame({
    "Emp-ID" : ["E101", "E102", "103", "E104"],
    "Salary" : ["25000", "$60000", None, "abc"]
})
df["Emp-ID"] = df["Emp-ID"].apply(
    lambda x : x if str(x).startswith(
        "E") else "Invalid")
df["Salary"] = df["Salary"].replace(
    ["$"], "", regex = True)
df["Salary"] = pd.to_numeric(df["Salary"],
    errors = "coerce")
df["Salary"] = df["Salary"].fillna(
    (df["Salary"].mean()))
print(df)
```

3. STUDENT PERFORMANCE ANALYSIS :

```
import pandas as pd
df = pd.DataFrame({
    "Roll-no" : [1, 2, 2, 3],
    "Marks" : [85, None, None, 70],
    "Grade" : ["A+", "Excellent", "Excellent", "B"]
})
df = df.drop_duplicates()
```



```
df["Marks"] = df["Marks"].fillna(df["Marks"].mean())
grade_map = {"Excellent": "A+", "A+": "A+", "B": "B"}
df["Grade"] = df["Grade"].map(grade_map)
print(df)
```

4. E-COMMERCE SALES DATASET PREPARATION:

import pandas as pd

```
df = pd.DataFrame({
    "Order-ID": [101, 102, 102],
    "Category": ["Electronics", None, None],
    "Order-Date": ["01-01-2024", "2024/02/01", "2024/02/01"]
})
```

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```
df = df.drop_duplicates()
```

```
df["Category"] = df["category"].fillna("Unknown")
```

```
df["Order-Date"] = pd.to_datetime(
    df["Order-Date"].dt.strftime("%Y-%m-%d")
)
```

```
print(df)
```

5. HOSPITAL PATIENT RECORDS INTEGRATION:

fillna(df
mean())
"A+" = "A+",

map(grade
map)

ET

None]

2024 /
02/01]

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import pandas as pd

registration = pd.DataFrame({

"Patient-ID": [1, 2], "Name": ["John", "Alice"]

})

laboratory = pd.DataFrame({

"Patient-ID": [1, 2], "Test": ["Blood", "X-Ray"]

})

billing = pd.DataFrame({

"Patient-ID": [1, 2], "Bill": [2000, 3500]})

df = registration.merge(laboratory, on =

"Patient-ID")

df = df.merge(billing, on = "Patient-ID")

df = df.drop_duplicates()

print(df)

6. SOCIAL MEDIA ANALYTICS DATASET:

import pandas as pd

df = pd.DataFrame({

"User-ID": ["U1", "U2", "U3"],

"Likes": [100, None, 200],

"Comments": [20, 15, None]})

df["User-ID"] = df["User-ID"].str.upper()

df["Likes"] = df["Likes"].fillna(0)

df["Comments"] = df["Comments"].fillna(0)

print(df)

7. BANKING TRANSACTION DATA CLEANING

unpaired pandas as pd

```
df = pd.DataFrame({
    "Acc-no" : [101, None, 103],
    "Amount" : [5000, -100, 3000],
    "Date" : ["01/01/24", "2024-02-01",
              "03-01-2024"]
})
```

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df ["Acc - no"] = df ["Acc - No "]. fillna
("Unknown")

$$df = df[df["Amount"] > 0]$$

df["Date"] = pd.to_datetime(df["Date"] - dt.strptime("-", "%Y-%m-%d"))

```
print (df)
```

8. IOT SENSOR DATA PROCESSING:

import pandas as pd

```
df = pd.DataFrame({
```

$$df["Temp"] = df["Temp"] - fillna(df["Temp"] - mean())$$

$df["Pressure"] = df["Pressure"].fillna(df["Pressure"].mean())$


```

df = df[df["Temp"] < 100]
df["Timestamp"] = pd.to_datetime
(df["Timestamp"] - dt.datetime(
    "1.1.1" - "1.1.1"
))
print(df)

```

9. HEALTH CARE COVID-19 DATA SET PREPARATION :

```

import pandas as pd
df = pd.DataFrame({ "Pati-ID" : [1, 2, 2],
                    "District" : ["Chennai", "chennai", "chennai"],
                    "Age" : [25, -5, -5] })
df = df.drop_duplicates()
df["District"] = df["District"].str.title()
df.loc[df["Age"] < 0, "Age"] = None
df["Age"] = df["Age"].fillna(df["Age"].mean())
print(df)

```

10. DATA WRANGLING FOR ML

```

import pandas as pd
df = pd.DataFrame({ "Cust-ID" : [1, 2, 2],
                    "Age" : [25, None, None],
                    "Gender" : ["Male", "Female", "Female"],
                    "Country" : ["India", "india", "india"],
                    "Unisex" : ["X", "Y", "Y"] })
df = df.drop_duplicates()
df["Age"] = df["Age"].fillna(df["Age"].mean())

```



```
df["Courtney"] = df["Courtney"].str.title()
df = pd.get_dummies(df, columns=["Gender"])
df = df.drop(columns=["Unused"])
print(df)
```